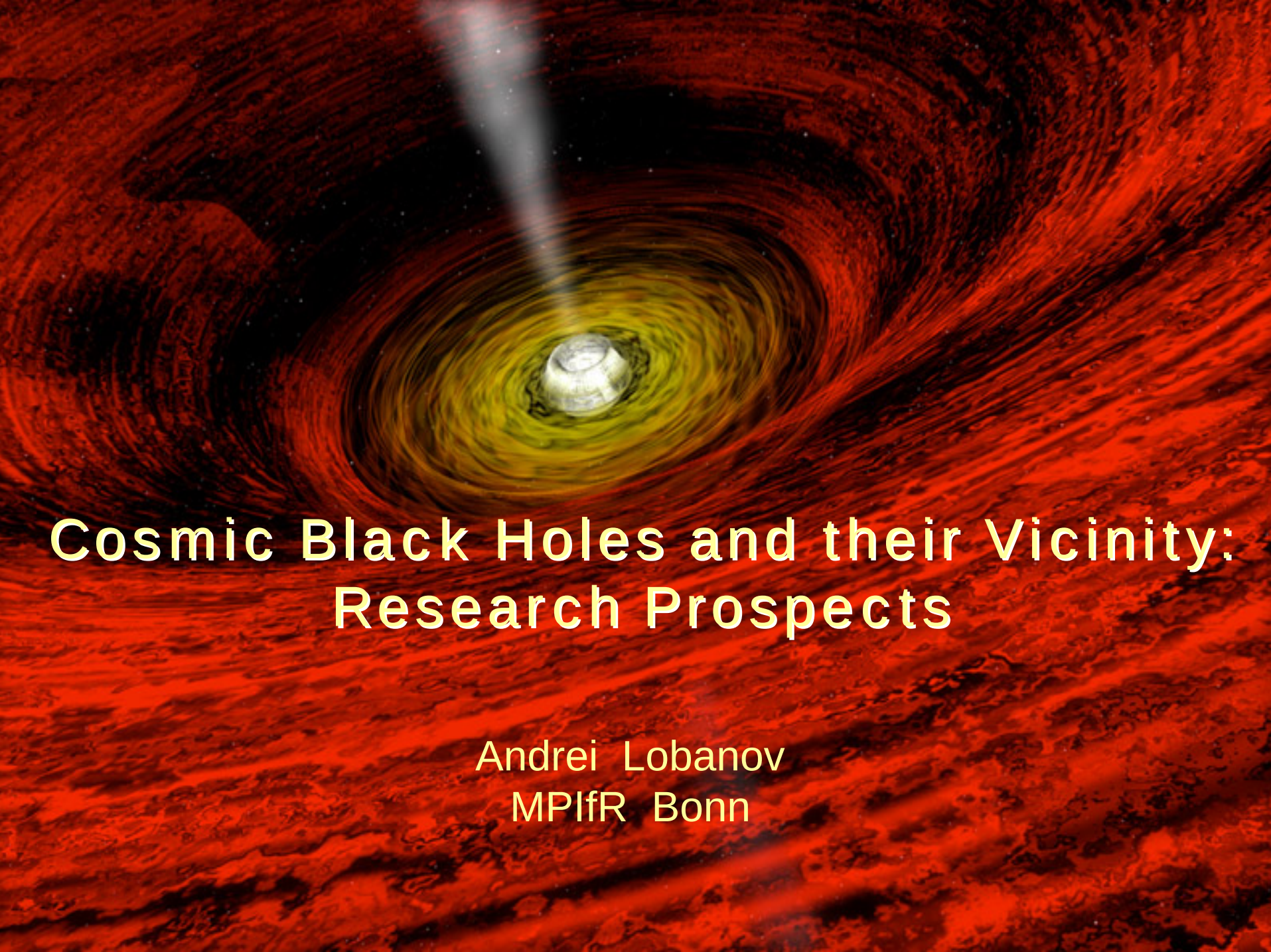

Cosmic Black Holes and their Vicinity: Research Prospects

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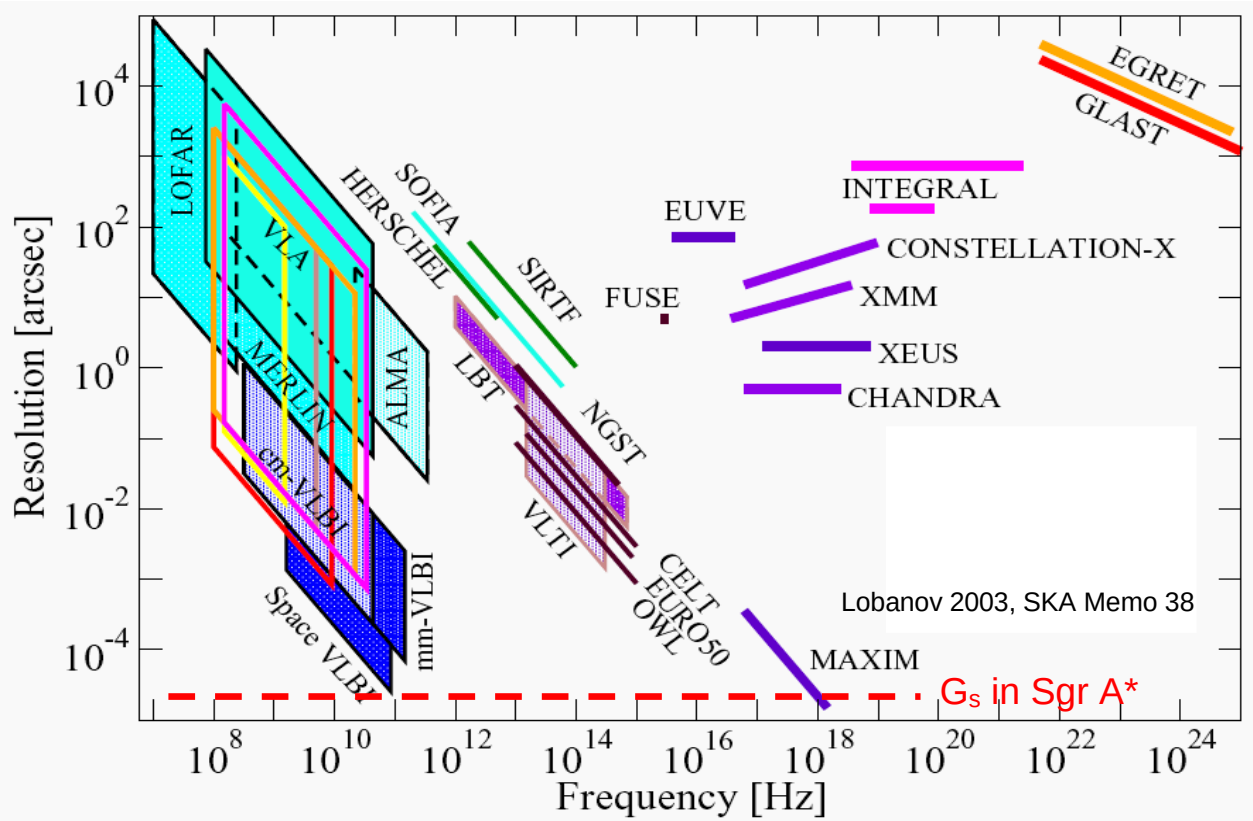
Opening Remarks

Thinking in terms of the next 20 years of BH research, we should:

- **Go beyond post-Newtonian tests of GR.**
- **Consider carefully all „exotic“ alternatives to BH (boson or quark stars, ECO, etc., including something totally unknown).**
- **Strive for a „full“ (*3D GRMHD + Poynting flux + thermal particles*) description of the BH vicinity (*relativistic jets and accretion disks*).**
- **Look for means to obtain 2D information about the BH vicinity (*true* imaging).**
- **Be realistic (or even opportunistic) in the choice of tools.**

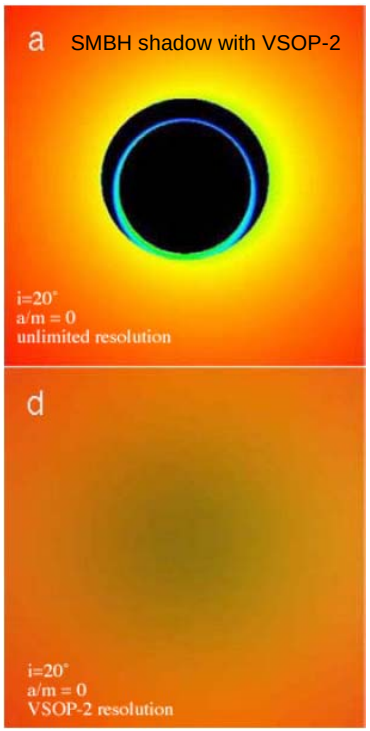
Studying the BH Vicinity

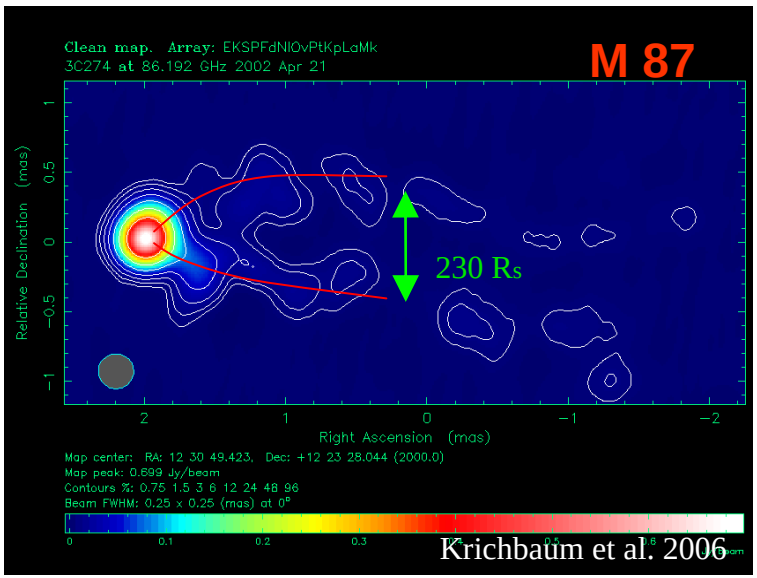
- **X-ray:** spectroscopy (1D, model dependent); interferometry (not available)
- **Optical, IR:** interferometry (need good uv-coverage, phase closures)
- **GWave:** interferometry (detection depends on pre-calculated templates)
- **Radio:** GVLBI (2D, calibration), SVLBI (2D, orbit determination)

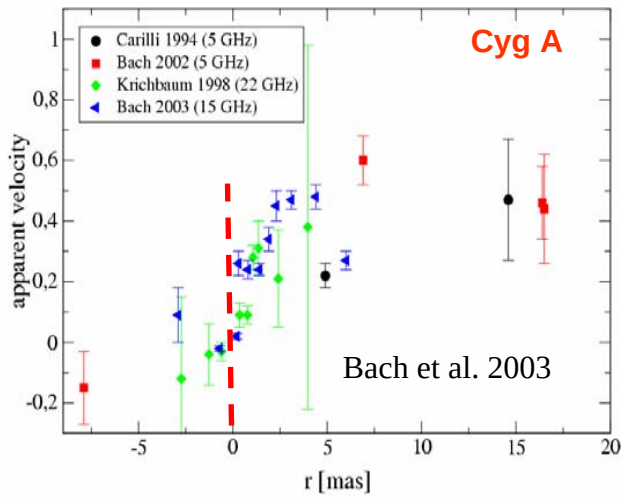


VLBI on BH Vicinity

- Direct imaging of hot (10^9–10^{10} K) material in AGN accretion disks in the vicinity of SMBH (M87: a BH "shadow" size ~26 μas, VSOP-2: ~40 μas; mm-VLBI: 20 μas @ 215 GHz).
- Formation, acceleration, collimation and internal structure of relativistic jets.

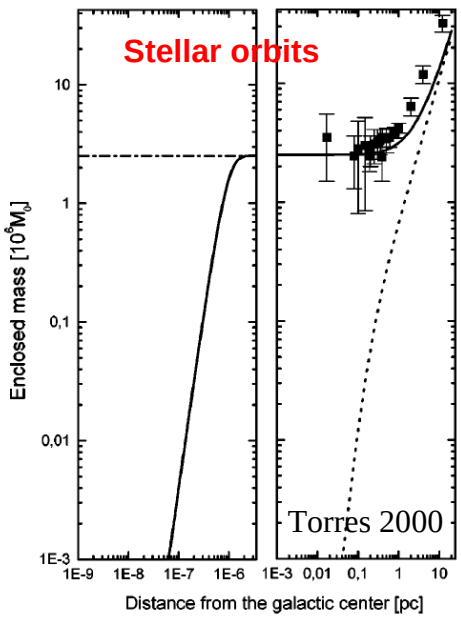


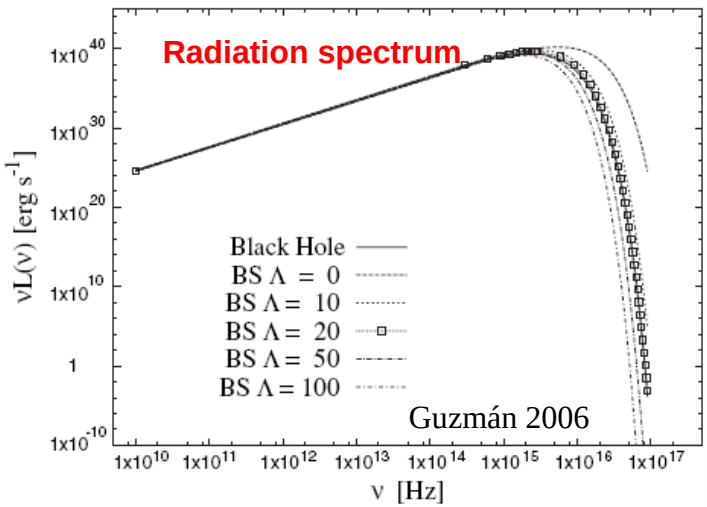


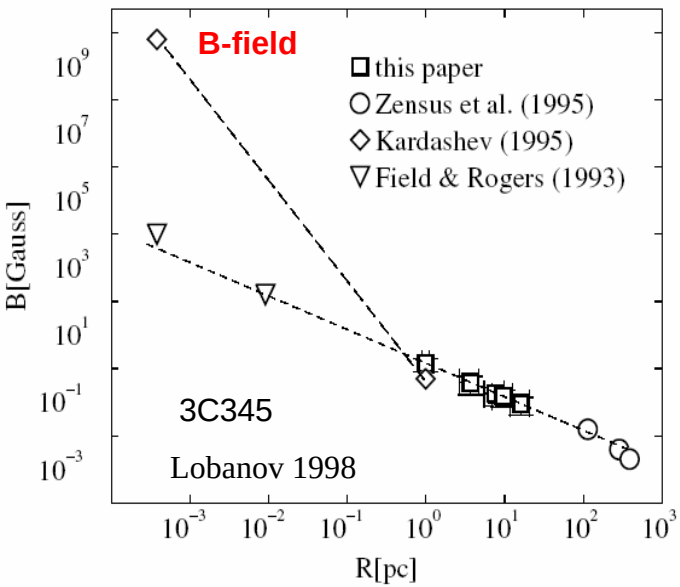


BH or not BH?

- Present evidence does not strictly prove existence of BH.
- Need to devise instruments and experiments to distinguish effectively between BH and their alternatives:
  - **stellar orbits:** (S1, Sgr A*) good enough for BH vs. ν condensate tests
  - **radiation spectrum:** high energies (BH vs. BS), ELF (BH vs. MECO)
  - **gravitation waves:** BH vs. anything (but need accurate templates)
  - **VLBI:** 2D imaging (BH vs. BS/MECO?), B-field (BH vs MECO)

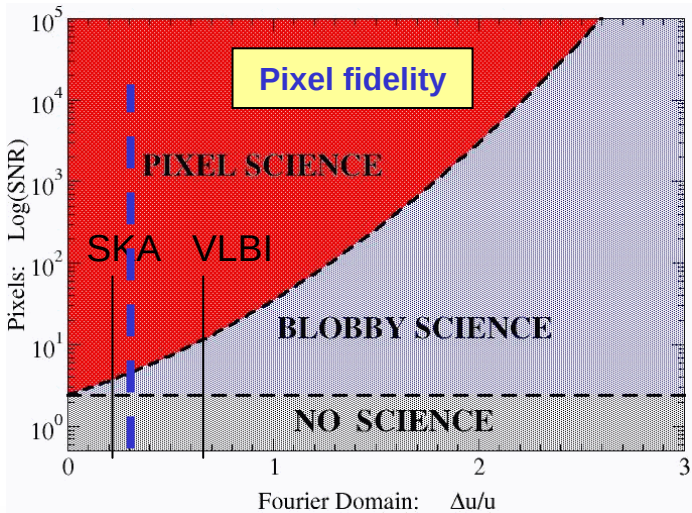


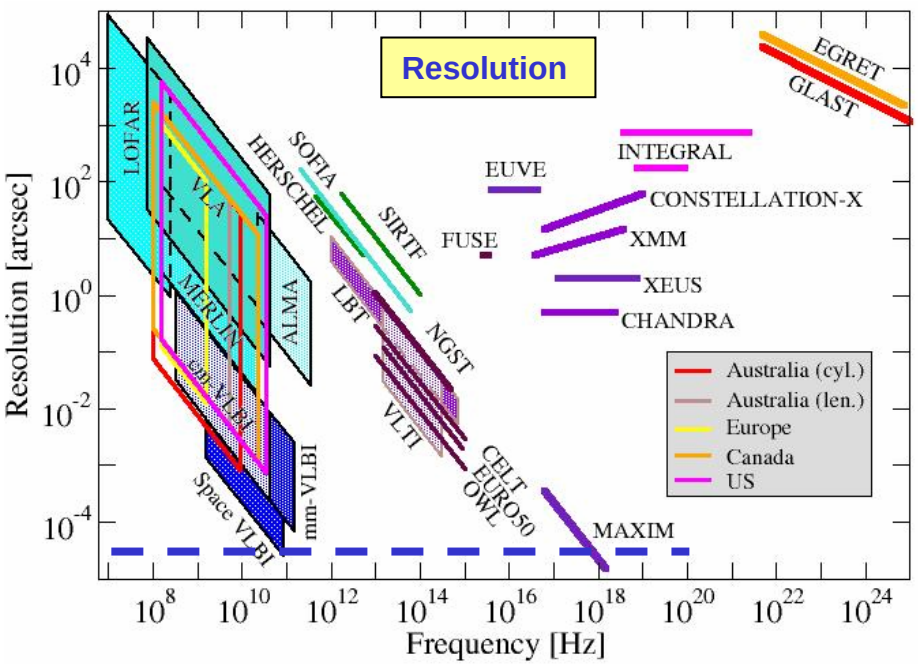


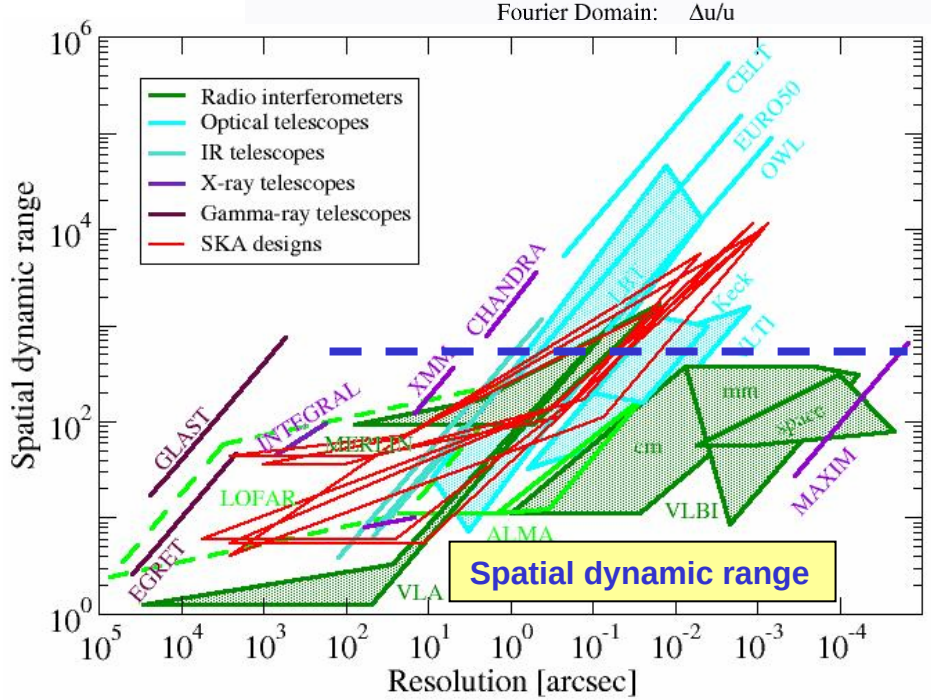


Observational requirements

- Effective studies of SMBH and their vicinity will require imaging with:
  - ~10 μas resolution;
  - spatial dynamic range of >500
  - pixel fidelity down to SNR~5

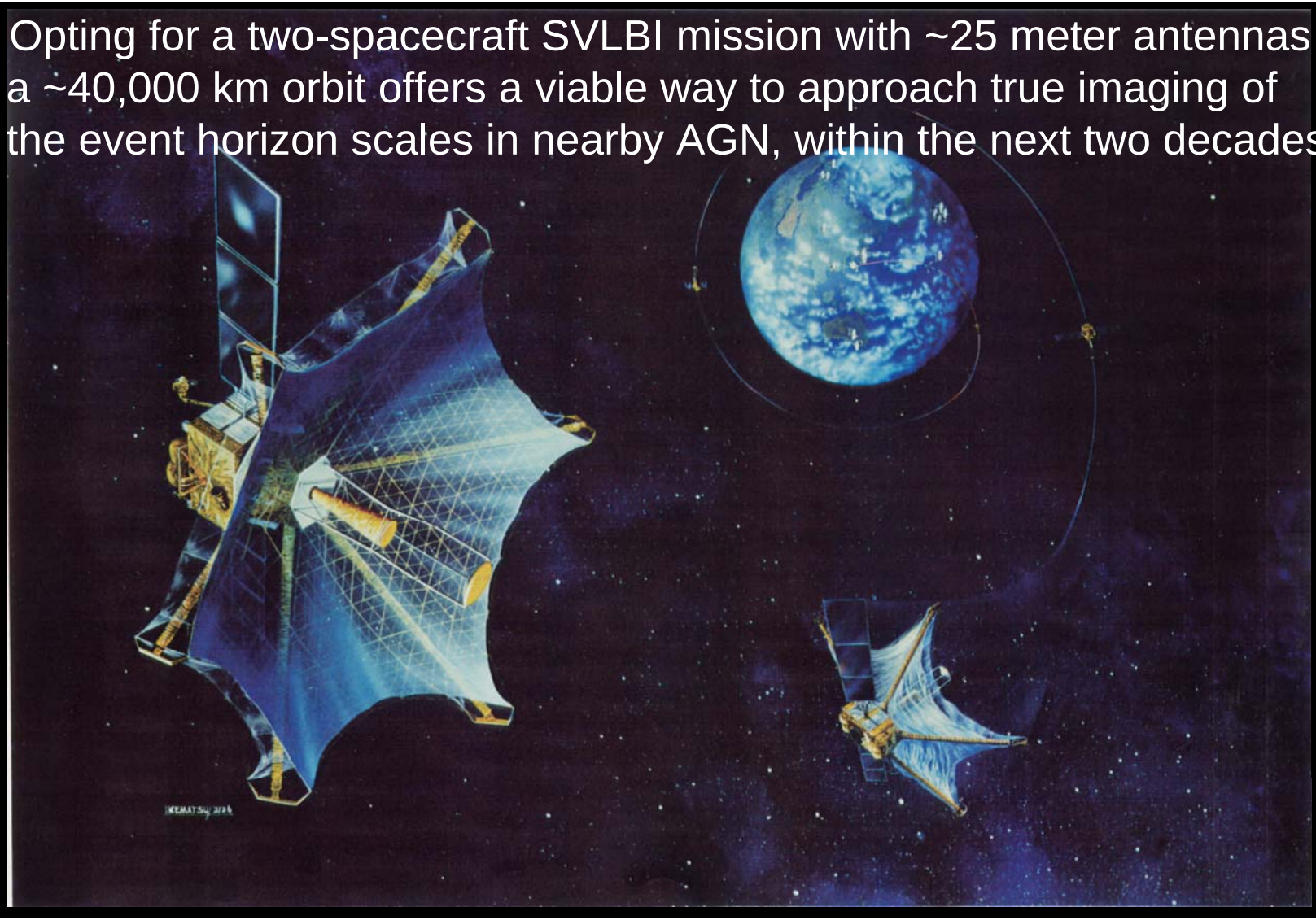






Space VLBI Prospects

- Opting for a two-spacecraft SVLBI mission with ~25 meter antennas at a ~40,000 km orbit offers a viable way to approach true imaging of the event horizon scales in nearby AGN, within the next two decades.



Summary

- Determining the true physical nature of black holes is fundamental for astronomy and physics.

- Jets may have a substantial EM component arising from an extreme vicinity of BH and affecting their formation and propagation. A 3D GRMHD description and high resolution imaging are essential for future studies.

- High-sensitivity VLBI from the ground (mm-VLBI) and space (VSOP-2 and beyond) is one of the primary (and affordable) tools for studies of BH and relativistic jets.